Plan — Object-Generic Geometric Approach Layer

Explicit closed-form branch-continuous planar IK transit `HOME`→`READY` (no RL, no BC)

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Abstract

The stricter compositional benchmark starts the arms at a collision-free home posture and asks the frozen learned stack to deliver strict K6. The audit (`HOME_START_COMPOSITION_NEEDS_UPSTREAM_REACH_SKILL`) showed the frozen `APPROACH` has no reach-to-contact capability, so a *single* upstream skill is missing: `HOME`→pre-capture `READY`. The key realisation of this plan is that for a **planar 2R arm this upstream skill is not a learning problem at all** — the inverse kinematics has an exact closed form (Siciliano), with exactly two elbow branches selected by the sign of q_2. We therefore build the reach as an **object-generic Geometric Approach layer**: object geometry → approach side → pre-contact tip targets $p^\star = p_{\text{contact}} - d_{\text{pre}}\,\hat{n}$ → analytic branch-continuous IK along a collision-free tip-space path → `READY`, time-parameterised under the frozen slew/torque contract. Learning is reserved for what kinematics cannot cover — the contact-rich *dynamic capture* (velocity-matched straddle for a coin) and the existing R2 transport. A learning-free positive control is already in hand: from the retained generic home `V1` the analytic transit reaches the pre-capture shell with **coin perturbation 0.0 mm, zero contact, single elbow branch, slew-feasible (max joint step** $0.069 \ll 0.300$**)**. This plan formalises that probe into an audited, tested module and its contracts. *Kinematics where kinematics suffices; learning only at the contact.*

1 Scope and goal

Goal. A deterministic, auditable `HOME`→`READY` transit for the two-arm planar gripper, built entirely from explicit closed-form 2R IK + a collision-free tip-space path, with *no RL and no BC*. **Architecture (the reusable layer boundary):**

GEOMETRIC APPROACH (explicit, this change) → READY / pre-contact manifold → learned local contact skill (dynamic capture) → frozen APPROACH → HANDOFF_RESET → R2 → K6.

Non-goals (explicit stop conditions). No RL and no BC in this change (they are justified only for amortising the planner across many geometries/obstacles, which the current fixed planar problem does not need — analytic IK is more precise, simpler, and auditable). No modification of the frozen downstream (APPROACH / HANDOFF_RESET / R2 / coast / K6 / physics / safety). No dynamic-capture CEM and no TD3 in this change — they are the *next* milestones, committed separately.

2 The frozen contract (must hold at every step)

- Physics + motion contract unchanged (`env/motion_contract.py`: governed $\Delta\tau$, slew cap = `tau_rate`$\cdot$ `control_dt` ≈ 0.30, joint/coin speed hard caps). Every torque enters through the existing `step_ablation`→governor stack; no raw actuator writes, no teleport, no coin-state edit, no hidden force in deployment. State-editing appears *only* as an offline characterisation/calibration probe, never in the deployed transit.
- K6 monitor and the entire learned downstream (`HANDOFF_RESET`, R2 $\alpha = 0.15$, release/coast) unchanged and imported *as-is*; `git diff` on those modules must be empty.

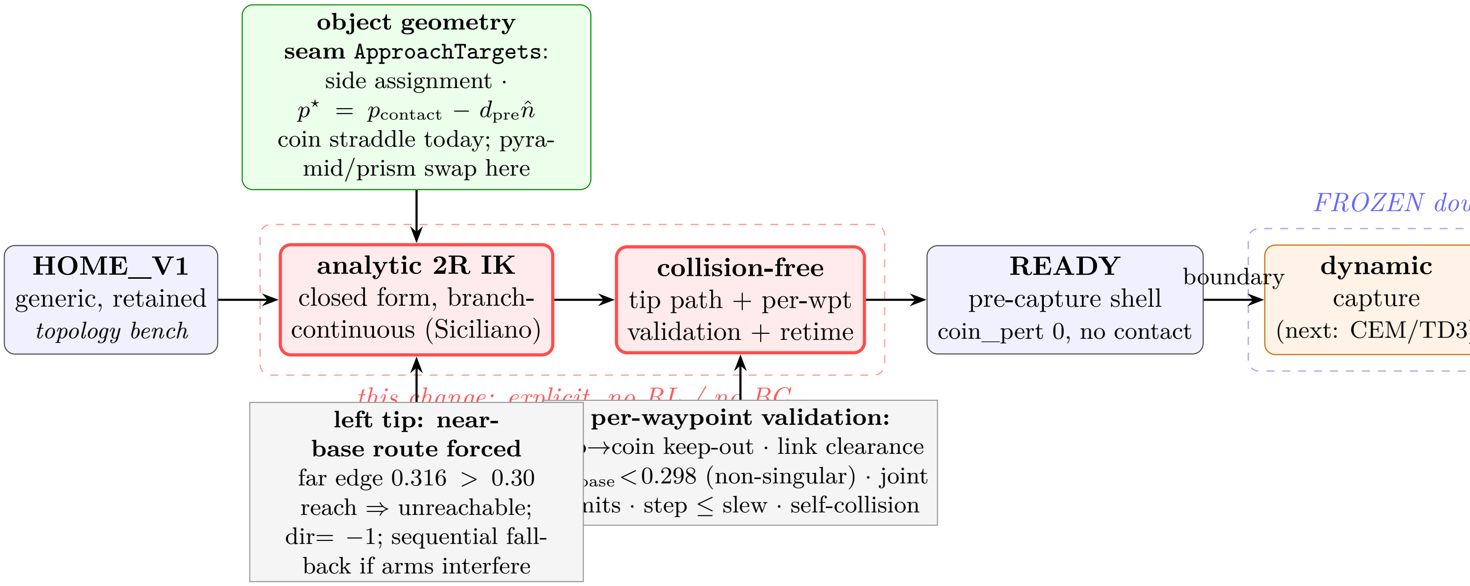

Figure 1: The object-generic Geometric Approach layer (red, this change) transits `HOME`→`READY` by explicit analytic IK over a collision-free tip path; everything downstream (blue frozen, orange learned-elsewhere) is untouched. The object-geometry seam (green) makes the pre-contact targets swappable per object (coin straddle today).

- `HOME_STATE_V1` is *retained verbatim* (`[-0.9,-1.4,0.0,2.7]`, committed `6d62b53e`); it is documented, not overwritten. `HOME_STATE_V2_READY` is a *new* sibling contract, not a redefinition of V1.
- `s4`/`s7` validation-only; `f1--f4 SEALED`; not opened by this change.

3 Affected files

CORE.YAML items touched: none (the coin RL work is entirely outside `hymeko_core`/`hymeko_query`/`hymeko_client`/`hymeko_daemon`/`parser` and touches no pinned dependency). All new code lives under `hymeko_rl/` + `tests/`/`reports/`/`docs/plans/`. **This change — new files only:**

- `hymeko_rl/coin_delivery/theta_option/planar_geometric_approach.py` — the layer: `PlanarArm2R` (calibrated closed-form IK), the `ApproachTargets` protocol + `CoinStraddleTargets` (object-geometry seam), `plan_collision_free_transit` (tip-space arc path + per-waypoint validation + sequential fallback + time-parameterisation), `GeometricTransitController` (rolls the reference through the governed servo).
- `hymeko_rl/coin_delivery/theta_option/home_states.py` — `HomeState` dataclass + `HOME_STATE_V1_GENERIC` and `HOME_STATE_V2_READY` contracts + `build_home_snapshot` (single source; the ad-hoc `build_home_state` in the composition experiment re-exports from here).
- `hymeko_rl/experiments/coin_kinetic_geometric_transit.py` — the transit gates + provenance JSON.
- `hymeko_rl/tests/test_planar_geometric_approach.py` — unit + integration + a lightweight perf assert.
- `reports/2026-07-28-planar-geometric-approach-transit.md` — the report.

Read-only reuse: `forward_displacement.py` (`_fingertip_geoms`, `_coin_xy`, `primary_fingertip_contacts`), `theta_option/kinetic_contract.py` (`TransportSnapshot.from_live`), `coin_strict_markov_ablation.py` (`step_ablation`), `env/motion_contract.py` (`govern_torque`), `teacher_bank.py` (`acquire_snapshot`, `load_harness`).
Later (not built here): `coin_kinetic_dynamic_capture.py` (short receding CEM from READY), then

a TD3 capture.

4 Interface changes (signatures, contracts)

All additive; no existing signature changes.

- `PlanarArm2R(shoulder_dof, elbow_dof, base, tip_geom, l1, l2)` with `calibrate(model, snapshot)` measuring $(\theta_b, s_0, o_1, s_1)$ from FK geometry. `ik(tip, branch)`$\to$`(a,b)` exact closed form; `ik_cont (tip, prev)` picks the branch minimising $\|\mathrm{wrap}(q - q_{\mathrm{prev}})\|$. **Pre:** `tip` inside the annulus $(|l_1 - l_2|, l_1 + l_2)$. **Post:** $\mathrm{FK}(\texttt{ik}(t)) = t$ to $< 10^{-3}\,\mathrm{m}$ (round-trip, asserted); the returned branch has constant $\mathrm{sign}(b)$ along a continuous in-annulus path.
- `ApproachTargets` protocol: `precontact(model, snapshot)`$\to$`(tipL*, tipR*, branchL, branchR)`. `CoinStraddleTarg` is the coin instance (two tips at the shell radius on the assigned sides). The seam is what a pyramid/prism instance would replace — *no* coin logic elsewhere.
- `plan_collision_free_transit(home, targets, arm_L, arm_R, cfg)`$\to$`TransitPlan` — builds each arm's tip path (radial-out $\to$ keep-out arc $\to$ radial-in to the assigned side), solves branch-continuous IK per waypoint, **validates every interpolated state** (tip$\to$coin keep-out, link$\to$coin clearance, in-annulus / non-singular $r_{\mathrm{base}} < r_{\mathrm{max}}$, joint limits, per-step $\leq$ slew), and returns a densified, slew-feasible `q_ref` trajectory. On a per-arm infeasibility it retries the alternative arc direction, then falls back to *sequential* (right-to-shell then left) execution; it never silently truncates — an unroutable request raises `TransitInfeasible` with the failing waypoint/reason.
- `GeometricTransitController.roll(home)`$\to$`TransitResult` — governed-servo tracking of `q_ref`; reports reached-shell error, coin perturbation, min tip$\to$coin clearance, contact count, peak qvel, and the terminal `TransportSnapshot` that the READY $\to$ capture boundary consumes. **Post:** deterministic given `(home, plan)`.

5 Test strategy (per layer)

Unit (`test_planar_geometric_approach.py`, deterministic seeds): calibration recovers $(\theta_b, s_0, o_1, s_1)$ and round-trips V1/V2/shell to 0 mm; `ik_cont` maintains elbow sign across a fine arc (a regression that would fail for a naive numerical DLS that hops branches); `ik` raises/clamps outside the annulus; `plan_collision_free_transit` *rejects* a path forced over the unreachable far side (`TransitInfeasible`) and *accepts* the near-base route; the produced `q_ref` has max step < slew; `CoinStraddleTargets` puts the two targets on opposite assigned sides at the shell radius. **Integration:** the full V1$\to$READY roll satisfies the four transit gates below on the canonical s1 coin. **Regression:** a coarse (corner-cutting) path must fail `COIN_UNPERTURBED` (the 100 mm-knock failure mode we measured), proving the densification/validation is load-bearing. **Boundary:** `REACH_TO_APPROACH_ONE_STEP_IDENTITY` — the terminal READY snapshot, advanced one online step, equals the offline READY state (no double phase-machine invocation, the boundary lesson); and an off-by-one mis-timed boundary is *rejected.* **Coverage:** every new public/private function exercised by ≥ 1 new test.

6 Performance budget

Measured anchors (this host, Apple-Silicon, `.venv` py3.11.15, mujoco 3.10.0): the full transit probe (151 waypoints $\times$ 3 substeps + hold) runs in < 1 s at RSS < 0.3 GB; calibration is ~8 FK evals. **Budget:** unit+integration+boundary suite < 90 s wall; **peak RSS < 1 GB** (hard cap 16 GB, never approached). The perf test asserts a numerical wall bound (< 3 s median over 5 iterations) and RSS bound on the transit roll.

7 Risk anticipation

Performance-contract preservation. The transit adds *no* new torque authority: `q_ref` is tracked by the *same* `step_ablation`→`govern_torque` stack the rest of the system uses, so the slew, torque, and joint/coin-speed caps are inherited unchanged. The densification guarantees per-step $\leq$ slew *before* tracking, so the servo never has to cut a corner into the coin. **Worst-case input / production scale.** The load-bearing geometric fact: the coin centre sits 0.271 m from each arm base while the arm reach is $l_1 + l_2 = 0.30$ m, so the coin's *far* edge ($0.271 + r_{\text{keepout}} \approx 0.316 > 0.30$) is *unreachable*. The left tip therefore *must* route the near-base way (dir $= -1$); the naive over-the-top arc drives IK onto the singular circle ($r_{\text{base}} \to 0.30$, $c \to \pm 1$) and the tracker then knocks the coin ($\sim$100 mm, measured). **Mitigation:** the planner validates $r_{\text{base}} < r_{\text{max}} = 0.298$ at every waypoint and rejects the far-side route, and the near-base arc is verified to stay non-singular (elbow bend $\gtrsim 20°$ at $r_{\text{base}} = 0.291$). The near-base swing must also clear the opposite arm and its links (self-collision check per waypoint) — the sequential fallback exists precisely for the case where the two arms' simultaneous near-base swings would interfere. **V2 singularity note.** `HOME_STATE_V2_READY`'s left tip sits at $r_{\text{base}} = 0.300$ (exactly full extension, $q_1 \approx 0$). V2 is used as a *ready-set target region*, not a hard pose to track through the singular point; the transit approaches READY *radially inward* (away from the singularity), and the READY guard tolerances (not a single pose) absorb the last millimetres.

8 Rollback path

New files only; rollback = delete them (no existing behaviour changes; `HOME_STATE_V1` untouched). The capture-CEM and TD3 milestones are separate later commits, each gated on its own test; a failure reverts to this transit without touching the frozen scaffold.

9 Gate ladder

1. `SINGLE_BRANCH_MAINTAINED_PASS`: the analytic IK holds one elbow sign per arm across the whole path (no topological switch; the far side is unreachable so the near-base branch is forced and consistent).
2. `COIN_UNPERTURBED_DURING_TRANSIT_PASS`: coin displacement $\leq \varepsilon$ (≈ 1 mm) and zero fingertip contact across the roll.
3. `TRANSIT_REACHES_READY_PASS`: both tips arrive inside the READY / pre-capture band (shell radius $\pm$ tol), qvel small.
4. `HOME_V1_TO_READY_COLLISION_FREE_PASS` *(this change's headline gate)*: all of the above hold on the retained generic `V1`, with per-waypoint validation and no state edit.
5. `REACH_TO_APPROACH_ONE_STEP_IDENTITY_PASS` *(boundary)*: offline READY = one online step from READY; an off-by-one is rejected.
6. *(Next milestones, not this change)* `HOME_V1_TO_DYNAMIC_HANDOFF_REACHABILITY_PASS` (≥ 3 planner seeds, short capture CEM from READY reaches H_{dyn}), then `HOME_V1_START_STRICT_K6_PASS` (TD3 finishes; teacher-free). If the capture reaches H_{dyn} but K6 fails, **HALT** and correct the certificate — do not tune the frozen downstream.

10 Empty-plan-dir hygiene

If abandoned before the transit module lands, delete `docs/plans/2026-07-28-planar-geometric-approach-transit/` in the same session.